

Emmanouil Giannakis

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EDUCATION

University of Piraeus, Piraeus, Greece

Sept 2021 - Present

BSc in Digital Systems,

Expected Graduation: June 2025

Thesis under the supervision of the Institute of Nuclear and Particle Physics at NCSR Demokritos:

"Neutrino's Event Reconstruction with Graph Neural Networks in ANNIE" Significant Projects:

- Developed a Sentiment Analysis system utilizing widely adopted NLP techniques and machine learning models, focusing on optimizing accuracy, performance, and computational efficiency for scalable deployment.
- Designed and implemented a multi-agent system with dynamic task allocation, negotiation protocols, and performance evaluation in a collaboration-driven environment.

Lyceum of Ayia Paraskevi, Athens Greece

2019 - 2021

- Honor Roll Student, High School Diploma Grade: 19.0 (GPA 4.0)

Henry Hudson Elementary School, Vancouver, Canada

Sep 2013 - Jun 2015

Studied at an international school and achieved native-level proficiency in English.

EXPERIENCE

Head of Autonomous System, Prom Racing FSAE, NTUA, Athens, Greece

Sept 2024 - Present

Technical:

- Familiarized with sensor calibration and fusion algorithms, including integration of data from IMU, INS, and camera-LiDAR systems. Gained experience in techniques to improve sensor alignment and accuracy, contributing to robust multi-sensor systems for autonomous navigation and environmental understanding
- Designed and implemented custom datasets with augmentation for object detection and pose estimation tasks. Utilized state-of-the-art architectures and leveraged YOLO-Pose to unify both tasks. Applied knowledge distillation to optimize model performance, achieving a reduction in inference time and enhanced precision with range detection.

Managerial:

- Secured a LiDAR sensor sponsorship, achieving a key milestone pursued for over three years.
- Oversaw two sub-team recruitments, meticulously selecting the most suitable candidates through a two-step recruitment process and ensured their integration into the team, through technical and safety training.

Autonomous System Engineer, Prom Racing FSAE, NTUA, Athens, Greece

Oct 2023 - Sept 2024

- Initiated R&D to evaluate the viability of computer vision projects and their investment potential.
- Customized a path planning algorithm inspired by RRT (Rapidly Exploring Random Trees) to fit specific requirements, integrating it into the ROS2 framework for enhanced autonomous navigation and decision-making.
- Presented the team's perception pipeline at a ZEISS competition, achieving 4th place among 42 top teams worldwide.

Student Researcher, NCSR Demokritos, Athens, Greece

Sept 2024 - Present

- Designing, implementing, and fine-tuning models based on the GraphNet architecture to optimize performance for event reconstruction tasks. Leveraged techniques from similar experiments to ensure reliable accuracy and predictive viability. Actively analyzing results to guide model improvements and adapt to requirements.
- Acquired extensive hands-on experience with Linux systems, utilizing them for daily personal and professional tasks particularly in system navigation, shell scripting, and troubleshooting.
- Co-developed and presented an open-source tool for converting ROOT files to SQLite, Parquet, and H5 formats, adopted in collaboration with the Fermilab team.

SKILLS

Technical Skills: Computer Vision, Digital Image Processing, Machine Learning, Deep Learning, Sensor Fusion, Filtering and Calibration Algorithms, Environment Perception, Multi-Agent Systems, Probability Theory, Stochastic Processes, Model Evaluation, Data Engineering and Visualization, and Testing.

Interpersonal Skills: Teamwork, Communication, Project Management, Presentation, Public Speaking, Problem Solving.

Languages: English (Proficient), Greek (Native)

Programming and Tools: Linux, Git, Bash, ROS2, Python, C, C++, Java, HTML, MATLAB, SQL.

CERTIFICATIONS

Volunteering Assistant, HIAS AI Summer School, NCSR "DEMOKRITOS"

July 2024

Fundamentals of Deep Learning, NVIDIA

May 2024

Unmanned Aerial Systems Common Module, Hellenic Air Force

Apr 2024

Universal AI Course, NCSR "DEMOKRITOS"

July 2023